

# Lab Assignment 9: Breast Cancer Diagnosis from Imaging Features

CSAI2017P: Applied Machine Learning (Lab)

Experiment 12 (Unit VI) • CO2, CO4

**Instructor:** Dr. Tofik Ali**Due:** announced in the lab**Student Name:** \_\_\_\_\_**SAP ID:** \_\_\_\_\_**Batch:** \_\_\_\_\_**Lab quiz:** LQ9, after submission

## Objective

- Build a diagnostic classifier where a false negative is the dangerous error.
- Use cross-validation properly on a small dataset.
- Reduce dimensionality and measure what it costs.

## Datasets used in this assignment

### B3 (Breast Cancer Wisconsin Diagnostic).

Full descriptions, sources and loading snippets for all anchor datasets are in **Anchor-Datasets.pdf**, alongside this brief.

## Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

### Submit:

- Notebook: `AML_LA09_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

| Section      | Level  | Choice rule                       | Marks     |
|--------------|--------|-----------------------------------|-----------|
| A            | Easy   | Attempt any 2 of 3 (2 marks each) | 4         |
| B            | Medium | Attempt any 1 of 2 (4 marks each) | 4         |
| C            | Hard   | Compulsory                        | 2         |
| <b>Total</b> |        |                                   | <b>10</b> |

Course outcomes assessed by this assignment: **CO2, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

**Section A (Easy) — attempt any 2 of 3    ( $2 \times 2 = 4$ )****A1. Load and inspect** [2 marks]

*Dataset: B3.*

- (a) Load the data, report the shape and the malignant/benign split.
- (b) Explain in two lines where the 30 features come from – they are not raw pixels.
- (c) Report the three features most correlated with the diagnosis.

**A2. First classifier** [2 marks]

*Dataset: B3.*

- (a) Fit logistic regression in a scaling pipeline with a stratified 80/20 split.
- (b) Report accuracy, precision, recall and F1 on the malignant class.
- (c) Report the number of malignant cases the model missed.

**A3. Cross-validation** [2 marks]

*Dataset: B3.*

- (a) Run 10-fold stratified cross-validation and report mean and standard deviation of accuracy.
- (b) Compare with your single-split figure from A2.
- (c) State in two lines why the single split is unreliable on 569 rows.

**Section B (Medium) — attempt any 1 of 2    ( $1 \times 4 = 4$ )****B1. Optimising for recall** [4 marks]

*Dataset: B3.*

- (a) Train logistic regression, SVM and random forest, and report recall on the malignant class for each.
- (b) For your best model, lower the decision threshold until malignant recall reaches at least 0.99, and report the resulting precision and the number of false alarms.
- (c) In 6–8 lines, argue what recall a screening tool should target and what the false alarms cost.

**B2. PCA and the cost of compression** [4 marks]

*Dataset: B3.*

- (a) Apply PCA inside the pipeline retaining 2, 5, 10 and 20 components.
- (b) Report cross-validated accuracy for each, plus the variance explained.
- (c) Plot the first two components coloured by diagnosis and explain in 4–6 lines what PCA bought and what it cost.

## Section C (Hard) — compulsory (2)

### C1. Would you deploy it

[2 marks]

*Dataset: B3.*

- (a) State the recall, precision and dataset size of your best model.
- (b) In 4–6 lines, list two things you would need to know about how the data was collected before letting this model near a patient.

*End of Lab Assignment 9*

*Sources:* Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.